

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I'm a software developer with over 10 years of experience. My last three have been in a Senior Developer/Architect role. I like building intricate and invisible systems, allowing for increased throughput by the teams using the tooling that I've built. Most recently, I've built a CLI that allows our developers to roll their own environments, along with some other convenience features. I am comfortable in the cloud, specifically AWS, through the use of modern DevOps tooling such as EKS, Karpenter, Helm, and Terraform. I have devised, taught, and helped enforce modern observability standards within our department through the use of Datadog and similar tools. I enjoy working in a challenging environment where I get to learn things and deliver value.

Work Experience

Alberta Motor Association

SOFTWARE ARCHITECT

Edmonton, Alberta

Jan 2023 - Present

- Built a self-service developer platform solo that took environment provisioning from 2-3 weeks down to 2 hours (60x). Self-service create/destroy, database snapshots, and pull-from-prod, adopted by 15 people across 3 teams and used to spin up 120+ environments. It made production debugging routine: a cancellation bug nobody could reproduce got root-caused in days using snapshotted production data. (Python, Terraform, AWS, Jenkins, Docker)
- Led a 2-3 engineer rewrite of a legacy document generation system into containerized microservices on EKS. 4-5 services (with sidecars) on a 3-5 node cluster, deployed with Helm. Wired up Datadog metrics for failure rate, throughput, generation time, and storage time, replacing a system that had frequent outages. (Kubernetes, EKS, Docker, Helm, Datadog, C#)
- Built an event-driven pipeline that ingests an external partner's policy data: S3 ingestion, a Kubernetes CronJob, a translation service, and transactional calls to a composite API, with OAuth 2.0 auth and Datadog monitoring. (Python, Kubernetes, AWS S3, OAuth 2.0)
- Architected the infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, 99.9% uptime since launch, 2.8M insured trips and 76,637 invoices processed. Won the 2024 Guidewire Innovation Award (1 of 3 worldwide). (Python, AWS Lambda, Terraform, TypeScript)
- Own 4 REST APIs serving 5 internal teams and external partners. Designed the contracts and versioning, approve new endpoint additions, and maintain OpenAPI design standards in Confluence. (Python, REST, OpenAPI, TypeScript)
- Set technical standards for a 12-person team through code reviews, RFC authorship, and PR checklists / API design guides. Member of the Architecture Steering Committee. Help newer developers navigate legacy codebases and trained the team on TDD methodology.

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built and maintained backend services that ran medical AI algorithms and returned results to the platform. (TypeScript, Node.js, AWS)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework that reduced defects by 30% on subsequent projects. Adopted across the insurance engineering team (15 developers / all Guidewire projects) and still in use 4+ years later. (Java, Gosu)
- Led 5-6 person development team on Guidewire insurance platform using TDD methodology. Performed Scrum Master role. Delivered projects on time and budget with no critical defects post-launch. Code reviews and mentoring junior developers.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local Guidewire development environments. Trained 5-6 developers on TDD methodologies through hosted training sessions.

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET, heavily utilizing the MVC design pattern, with REST integration and tests written with NUnit and Moq. (C#, .NET, MVC, REST)

Lockheed Martin Canada

ASSOCIATE SOFTWARE ENGINEER

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above-water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging and testing capabilities.

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

Education

The Open University MSC. COMPUTING (SOFTWARE ENGINEERING) Currently pursuing Master's in Computing at The Open University.	Milton Keynes, England (Remote) May 2024 - Ongoing
The King's University B.S. IN COMPUTER SCIENCE	Edmonton, Alberta Sep 2012 - Dec 2016

Skills

Developer Platforms & Tooling	Built a self-service developer platform solo that cut environment provisioning from weeks to hours (60x), with create/destroy, database snapshots, and pull-from-prod. Adopted by 15 people across 3 teams, it spun up 120+ environments and turned production debugging that used to take weeks into something that takes hours. The kind of automation that makes other engineers faster is the work I gravitate toward.
Cloud-Native Infrastructure	Production AWS across multiple accounts (Lambda, EKS, SQS, RDS, IAM, ECR) with Terraform for infrastructure as code. Run a production EKS cluster with cert-manager, Karpenter, and Linkerd, and ship containerized microservices on it with Helm. Comfortable across CI/CD, Docker, and the pipelines that build and deploy all of it.
Reliability & Observability	Instrument services with Datadog metrics for failure rate, throughput, and latency, and use them to define what healthy looks like and alert when it isn't. I like chasing hard production failures to root cause: a cancellation bug nobody could reproduce got fixed in days once I could replay real production data against it.
Technical Leadership & Ownership	Set technical standards for a 12-person team through code reviews, RFC authorship, PR checklists, and API design guides. Member of AMA's Architecture Steering Committee. Drive cross-team architectural decisions and own technical domains from design through production. I write a lot, because clear docs are how direction actually sticks.

Technologies

Backend Languages	Python, TypeScript, Java, Node.js, Bash, C#
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM, ECR), Terraform, Kubernetes, Docker, Helm
CI/CD & Automation	CI/CD, GitHub Actions, Jenkins, Azure Artifacts, GitOps, Infrastructure as Code
Architecture & APIs	REST APIs, OpenAPI, Microservices, Event-Driven Architecture, Async Workflows
Databases	PostgreSQL, Aurora/RDS, Redis
Observability	Datadog, Elasticsearch, Kibana, Metrics Design, SLOs, Alerting